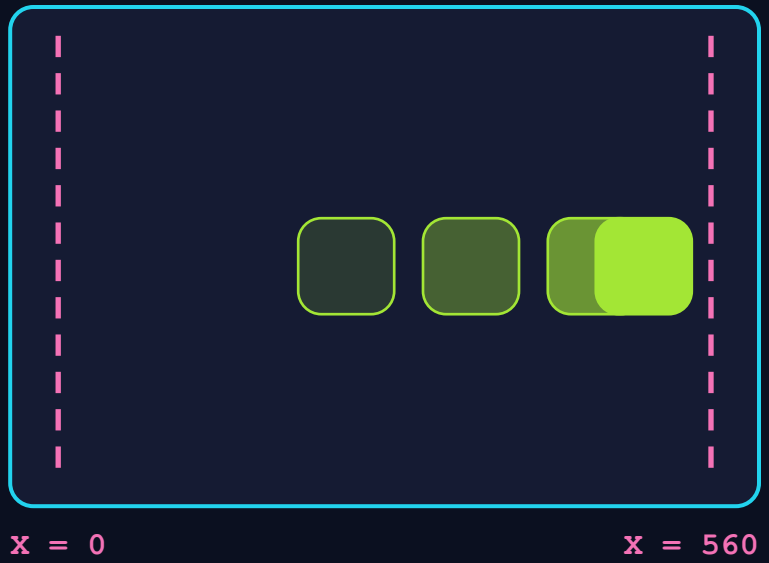
CCA GAME DEVELOPMENT

Invisible Walls

Module 2 — Session 7

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```
if player_x < 560:
```

Today's Goal

By the end of this session:

01



Understand how boundaries work

02



Use if statements to control movement

03



Keep the player inside the screen

04



Test and fix movement problems

Can You Escape the Screen?



Remember Session 6?

Press RIGHT → Player moves right

But what happens if we keep pressing RIGHT?

The player can move outside the screen!

Today's mission: Keep the player inside the game window.

Where Should the Player Stop?

Imagine our screen is 600 pixels wide. Our player is 40 pixels wide.



The player's position is its left edge, not its center.

Max X = Screen Width − Player Size

600 − 40 = 560

Minimum X = 0 · Maximum X = 560

Think: Why can't the player's X position reach 600?

Make a Decision Before Moving

An if statement checks a condition.

RIGHT BOUNDARY

```
if player_x < 560:
    player_x += 20
```

LEFT BOUNDARY

```
if player_x > 0:
    player_x -= 20
```

IF the player has not reached the boundary, **THEN** move.

IF → CHECK → ACTION

But there is one problem.
What if the player is at 550 and moves 20 pixels right? It reaches 570, beyond our boundary!
How can we fix this?

Move → Check → Correct

We can correct the player's position after moving.

RIGHT SIDE

```
if event.key == pygame.K_RIGHT:
    player_x += 20

    if player_x > 560:
        player_x = 560
```

LEFT SIDE

```
if event.key == pygame.K_LEFT:
    player_x -= 20

    if player_x < 0:
        player_x = 0
```

MOVE → **CHECK** → **CORRECT**

Even if the player moves too far, the code puts it back inside the boundary.

Mission 1: Build Invisible Walls

Open the Session 7 starter code
`https://github.com/annisafzhn/CCA.git`

Find the Session 6 project and locate the LEFT and RIGHT keyboard controls

Update the movement code to include boundaries

TEST YOUR GAME

- Press RIGHT repeatedly.
- Press LEFT repeatedly.
- Try changing the movement speed.
- Check whether the player stays inside the screen.

Success condition: The player can move freely but cannot escape the window.

Mission 2: Test Your Logic

Complete these challenges in order.

1 Predict and Test

The player is at X = 540. It moves right by 20 pixels. What will happen?

A. The player stops at 540

B. The player reaches 560

C. The player reaches 580

Run the game and check your answer.

2 Change the Speed

Change the movement speed from 20 to 50.

20 → 50

Test the right boundary.

Does the player stay inside the screen? Why?

3 Debug the Code

Find the problem:

```
if event.key == pygame.K_RIGHT:
    player_x += 20

if player_x > 560:
    player_x = 560
```

The boundary check is outside the keyboard event condition.

Think: When should the game check the player's position?

Test your fix and explain the result.

Tomorrow: Dodge the Block!

What will we build?

SESSION 5
X, Y Movement

+

SESSION 6
Keyboard Triggers

↓

SESSION 7
Boundaries

↓

SESSION 8
DODGE THE BLOCK!

YOUR MISSION

Move the player.

Stay inside the screen.

Avoid falling blocks.

Use what you've learned. Build something playable!

Today's Big Ideas

`BOUNDARY`	The limit of an object's movement.
`IF STATEMENT`	Checks whether a condition is true.
`SCREEN WIDTH - PLAYER SIZE`	Calculates the maximum X position.
`MOVE → CHECK → CORRECT`	A way to keep the player inside the screen.

NEXT SESSION: DODGE THE BLOCK

Module 2 — Session 8

Can you use everything you've learned to build a playable survival game?

See you in Session 8! 👋